PROG1350A - Python III: Data Preprocessing and Statistical Analysis

Assignment I (Group Assignment)

Data Cleaning and Variable Encoding with Pandas

Objective:

To prepare a churn dataset for analysis using the different Pandas functions learned in class. The expected output should be a cleaned, ready-to-use dataset for an analytical method.

Instructions:

This assignment is worth 15 points in total. To get 10 points, you must complete the required tasks listed below:

1. Identify numeric and categorical variables and their data types. Identify the number of columns and rows per file. ***(0.5 points***)
2. Identify the number/percent of missing data per column. Decide whether any columns should be dropped based on these results and drop them if necessary. Additionally, drop any columns that you consider irrelevant for the project. Assume that unique identifiers are irrelevant for your analysis. ***(1 points)***
3. Identify any data inconsistencies and fix them (e.g.: datatype conversion, string reformatting, invalid data, etc.). ***(2 points)***
4. Based on point 3, perform data inputting on the columns that require it. Optionally, you can remove rows with missing values. ***(2 points)***
5. Detect and remove duplicates if present. Identify potential outliers in your dataset and decide whether to keep or remove them. ***(2 points)***
6. Encode categorical variables using One-Hot encoding or Label encoding (choose the most suitable method for each of the variables). *Your resulting dataset should not include the original variables after the encoding process.* ***(2 points)***
7. Give a summary of your final dataset, including the number of rows and columns as well as the variables and datatypes. Include also a view of the first 5 records of the dataset. ***(0.5 points***)

The remaining **5 points** will be assigned according to your effort on preparing the dataset and explaining the rationale for each of your decisions and your code in a “technical interview” of

approximately 15 minutes (PowerPoint is not required, only Jupyter Notebook). Focus your explanations on tasks 2-6 (***1 point each***).

Deliverables:

- A Jupyter Notebook file containing the code and explanations of your data cleaning process.
- Code explanation (*all team members must participate, all members get the same mark*).
- The cleaned dataset (csv format)

Note: ***ALL*** *files need to be submitted to receive a mark. If a file is missing or does not compile (in case of the Jupyter Notebook file), the corresponding delay penalty will be applied (20% of the maximum value per day late).*

Due Date:

- September 28, 11:59 pm.

Teams:

- Groups of two or three.

Dataset Description:

A fictitious dataset about churn from a Telecom provider. The data can be classified into 4 categories:

a) Customer status

- Customer ID: The unique ID that identifies a customer
- Churn Label: Contains “Yes” or “No” to indicate if a customer churned
- Churn Category: Group multiple churn reasons together for analysis purposes
- Churn reason: The particular reason why the customer ended the contract

b) Customer demographics

- Gender: The gender of the customer, indicated by “Male”, “Female” or “Prefer not to say”

- Age: The age of the customer

- Under 30: Indicates if the customer is under 30 with “Yes” or “No”

- Senior: Indicates if the customer is above 65 with “Yes” or “No”

c) Contract information

- Contract Type: Contains “Month-to-Month”, “One year” or “Two Year”

- Payment Method: Preferred payment method of the customer indicated with “Credit Card”, “Direct Debit”, or “Paper Check”

- State: The code of the state where the customer lives

- Phone number: Phone number of the customer

- Group: Indicates if the customer is part of a group contract. A group contract offers advantages and is generally cheaper. Contains “Yes” or “No”.

- Number of customers in a group: Number of customers part of the group.

d) Subscription types and charges

- Account Length (in months). The number of months the customer has been with the company.

- Local calls: Amount of local (within the US) calls from the customer

- Local mins: The number of minutes spent calling internationally.

- Intl Active: Indicates if the customer called internationally with a “Yes” or “No”.

- Int Plan: Indicates if the customer has a premium plan to call internationally for free with “Yes” or “No”. This premium is reflected in the amount of the monthly charge.

- Extra International Charges: Contains the extra charges for international calls for customers who are not on an international plan.

- Customer Service Calls: The number of calls made to customer service.

- Avg Monthly GB Download: Contains the average monthly download volume in gigabytes.

- Unlimited Data Plan: Indicates if the customer has free unlimited download capacity with “Yes” or “No”. This premium is reflected in the amount of the monthly charge.

- Extra Data Charges: Contains the extra charges for data downloads for customers who are not on an unlimited plan.

- Device Protection & Online Backup: Average of all monthly charges to the customer.

- Total Charges: Sum of all monthly charges.

Data Files:

File: *customer_churn.xlxs*